$$\alpha = \sum_1^n \frac{1}{n}\delta_{x_i}, \beta = \sum_1^n \frac{1}{n}\delta_{y_j}$$

Particular Case: Uniform Measure

Consider: $\alpha = \sum_1^n \frac{1}{n}\delta_{x_i}$, $\beta = \sum_1^n \frac{1}{n}\delta_{y_j}$ (i.e. $n = m$, uniform marginals)

What are valid couplings in this case?

Bistochastic (aka doubly-stochastic) matrices: $\sum_i \pi_{ij} = \sum_j \pi_{ij} = \frac{1}{n}$

In this case, the transportation polytope $\Pi(\mathbf{1}, \mathbf{1})$ is called the **Birkhoff polytope,** denoted $\mathscr{B}_n$

Pop-Quiz

Consider the uniform case $\mathbf{a} = \frac{1}{n}\mathbf{1}_n, \mathbf{b} = \frac{1}{n}\mathbf{1}_n$.

Which one is larger? $\Pi(\mathbf{a}, \mathbf{b})$ or Perm(n)?